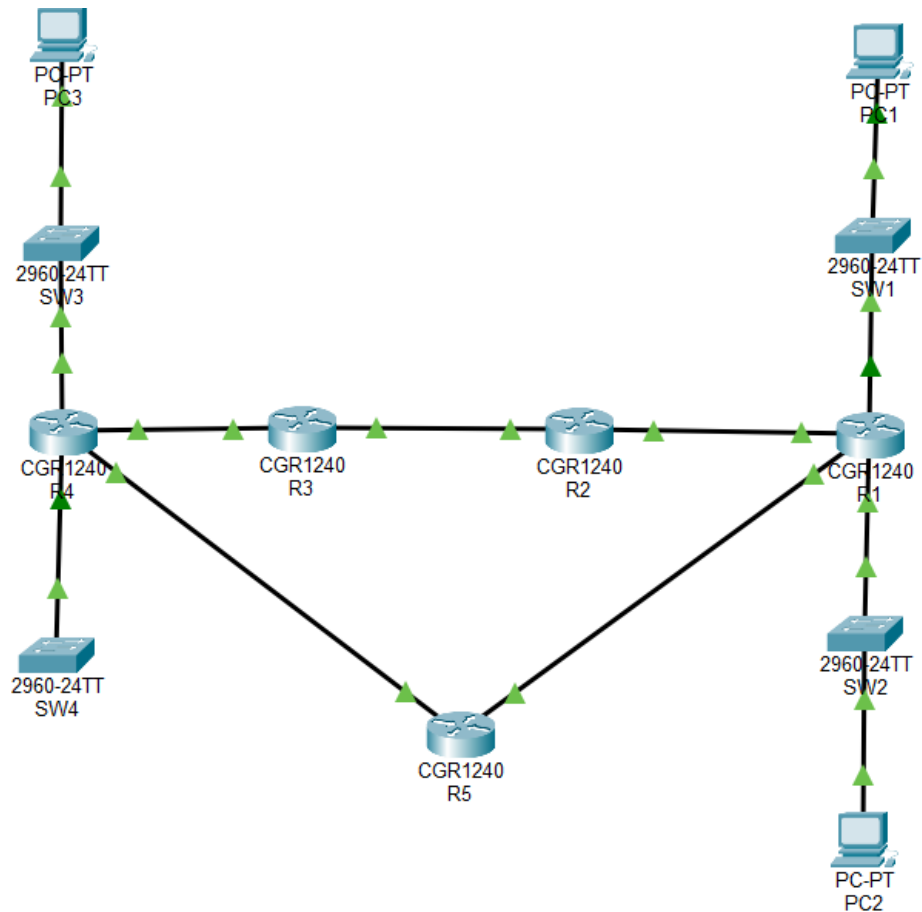


Week 8: Dynamic Routing Protocols (5 Marks)



74. R1:

```
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
```

R2:

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip routing
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#end
R2#
```

R3:

```
R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip routing
R3(config)#router rip
R3(config-router)#network 10.0.0.0
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
```

R4:

```
R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip routing
R4(config)#router rip
R4(config-router)#network 10.0.0.0
R4(config-router)#no auto-summary
R4(config-router)#end
R4#
```

R5:

```
R5>ena
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip routing
R5(config)#router rip
R5(config-router)#network 10.0.0.0
R5(config-router)#no auto-summary
R5(config-router)#end
```

75.

```
R1>ena
R1#debug ip rip
RIP protocol debugging is on
R1#
10.1.3.0 in 1 hops
RIP: sending v1 update to 255.255.255.255 via FastEthernet2/3 (10.0.0.1)
RIP: build update entries
    network 10.0.1.0 metric 1
    network 10.0.2.0 metric 1
    network 10.0.3.0 metric 1
    network 10.1.2.0 metric 3
    network 10.1.3.0 metric 2
```

76. R1:

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2:

```
R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#end
R2#
```

R3:

```
R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
%SYS-5-CONFIG I: Configured from console by console
```

R4:

```
R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#end
R4#
```

R5:

```
R5>ena
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#end
R5#
%SYS-5-CONFIG I: Configured from console by console
```

77.

```
R1>ena
R1#debug ip rip
RIP protocol debugging is on
R1#RIP: sending v2 update to 224.0.0.9 via FastEthernet2/3 (10.0.0.1)
RIP: build update entries
  10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
  10.1.2.0/24 via 0.0.0.0, metric 3, tag 0
  10.1.3.0/24 via 0.0.0.0, metric 2, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet2/4 (10.0.1.1)
RIP: build update entries
  10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
  10.1.2.0/24 via 0.0.0.0, metric 3, tag 0
  10.1.3.0/24 via 0.0.0.0, metric 2, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet2/5 (10.0.2.1)
RIP: build update entries
  10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
  10.1.2.0/24 via 0.0.0.0, metric 3, tag 0
  10.1.3.0/24 via 0.0.0.0, metric 2, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet2/6 (10.0.3.1)
RIP: build update entries
  10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
  10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
```

78.

```
R1#show ip route rip
      10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
R       10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:14, FastEthernet2/6
R       10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:14, FastEthernet2/6
```

79.

```
R1#show ip rip database
10.0.0.0/24    auto-summary
10.0.0.0/24    directly connected, FastEthernet2/3
10.0.1.0/24    auto-summary
10.0.1.0/24    directly connected, FastEthernet2/4
10.0.2.0/24    auto-summary
10.0.2.0/24    directly connected, FastEthernet2/5
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, FastEthernet2/6
10.1.2.0/24    auto-summary
10.1.2.0/24    [2] via 10.0.3.2, 00:00:05, FastEthernet2/6
10.1.3.0/24    auto-summary
10.1.3.0/24    [1] via 10.0.3.2, 00:00:05, FastEthernet2/6
```

80. R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 10.0.0.0 0.0.255.255 area 0
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2:

```
R2>ena
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)# network 10.0.0.0 0.0.255.255 area 0
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

R3:

```
R3>ena
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.0.0 0.0.255.255 area 0
R3(config-router)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R4:

```
R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#network 10.0.0.0 0.0.255.255 area 0
R4(config-router)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

R5:

```
R5>ena
R5# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router ospf 1
R5(config-router)#network 10.0.0.0 0.0.255.255 area 0
R5(config-router)#end
05:19:22: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.3.1 on FastEthernet2/4 from LOADING to FULL,
Loading Done
```

81:

```
R1>ena
R1#show ip route rip
      10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
R      10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:22, FastEthernet2/6
R      10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:22, FastEthernet2/6
```

82.

83.

```
R1#show ip ospf database
      OSPF Router with ID (10.0.3.1) (Process ID 1)

      Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
10.0.3.1     10.0.3.1     522         0x80000005  0x006c58  4
10.1.3.2     10.1.3.2     522         0x80000002  0x00bf57  1

      Net Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum
10.0.3.1     10.0.3.1     522         0x80000001  0x001a7d
R1#
```

84.

RIPv1 uses broadcast traffic and sends updates to 255.255.255.255. RIPv2 uses multicast traffic and sends updates to 224.0.0.9.

85.

There are two possible paths to the 10.1.1.0/24 network through different routers. RIP can learn the network through different paths, so more than one route may appear in the routing information before OSPF is configured.

86.

RIP routes may still exist in the RIP database, but OSPF routes are preferred in the routing table because OSPF has a lower administrative distance (110) than RIP (120). Only the preferred OSPF route to 10.1.1.0/24 is installed in the routing table

87.

R1's routing table changed because the path through R2 was no longer available. OSPF recalculated the network paths and selected an alternative path through the remaining available routers. The routing metric/cost and outgoing interface can therefore change, not just the next-hop address.

88.

The RIP database mainly contains learned networks, next-hop information and hop counts. The OSPF database contains detailed information about the network topology and link states between routers. This difference exists because RIP is a distance-vector protocol that learns routes from neighbours, while OSPF is a link-state protocol that builds a map of the network and calculates the shortest paths.